

DATA SCIENCE RESOURCES

Education Analytics Projects

Project 1. Student Dropout Predictor

Problem Statement: High dropout rates in universities and online learning platforms impact student success and institutional reputation. Early prediction of at-risk students allows for timely intervention.

Objective: Build a binary classification model to identify students likely to drop out based on demographics, attendance, academic performance, and behavioral data.

Dataset Source: [Open University Learning Analytics Dataset](#) (OULAD)

Techniques & Concepts

- Binary classification
- Feature engineering (engagement, activity counts)
- Handling class imbalance (SMOTE, class weights)
- Evaluation with ROC-AUC, precision, recall

Tools & Libraries: Python, Pandas, Scikit-learn, Matplotlib, XGBoost, Imbalanced-learn

Expected Outcome: A model that flags at-risk students for early outreach and support programs, with a dashboard view for academic counselors.

Project 2. Course Recommendation System

Problem Statement: Students often struggle to choose the right courses that match their interests, skills, or career goals.

Institutions need to personalize course suggestions to improve outcomes.

Objective: Build a recommender system that suggests relevant courses to students based on their past enrollments, preferences, or peer behavior.

Dataset Source: [edX Course Recommender Dataset](#)

Techniques & Concepts

- Collaborative filtering (user-item matrix, cosine similarity)
- Content-based filtering using course metadata
- Hybrid recommendation models
- Matrix factorization (SVD)

Tools & Libraries: Python, Pandas, Scikit-learn, Surprise, Streamlit (for interactive demo)

Expected Outcome: A personalized course recommender that displays top-k course suggestions per user.

Project 3. Exam Score Prediction Model

Problem Statement: Institutions want to proactively forecast exam performance to identify students who need academic support and to optimize learning resources.

Objective: Build a regression model that predicts exam scores based on attendance, assignments, quiz scores, demographics, and study time.

Dataset Source: [Student Performance Dataset \(UCI\)](#)

Techniques & Concepts

- Linear Regression, Decision Trees, XGBoost
- Feature scaling and encoding (categorical variables)
- Cross-validation and error analysis (MAE, RMSE)

Tools & Libraries: Python, Pandas, Scikit-learn, XGBoost, Seaborn, SHAP for feature importance

Expected Outcome: A regression model that predicts final exam scores with high accuracy and interpretable insights into key influencing factors.
